

Functions

- Functions are blocks of code designed to perform specific tasks.
- They can be reused and modularize code for better readability and maintenance.

2. Declaring Functions:

- Syntax:

javascript

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```
function functionName() {  
    // Code block  
}
```

- Functions are declared with the function keyword, followed by a name and a code block.

3. Calling Functions:

- Use the function name followed by parentheses:

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```
functionName();
```

- This executes the code inside the function.

4. Local Variables:

- Variables declared inside a function using let or var are local to that function.
- Example:

javascript

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```
function example() {  
    let localVar = 10;  
    console.log(localVar); // Accessible here  
}  
  
console.log(localVar); // Error: localVar is not defined
```

5. The return Statement:

- Terminates the function execution at the return point.
- Sends a value back to the caller.
- Example:

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```
function add(a, b) {  
    return a + b;  
}  
  
let sum = add(5, 10); // sum = 15
```

6. Function Parameters:

- Input values provided to functions for computation.
- Example:

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```
function greet(name) {  
    console.log(`Hello, ${name}!`);  
}  
  
greet("Stavan"); // Output: Hello, Stavan!
```

7. Shadowing:

- Local variables with the same name as global variables "shadow" the global ones inside the function.

8. Parameter Validation:

- Check the validity of parameters before using them:

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```
function divide(a, b) {  
    if (b === 0) {  
        return "Division by zero error!";  
    }  
}
```

```
    return a / b;
}
```

Example Code Breakdown:

Without Functions:

javascript

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```
let temperatures = [12, 13, 14];
let sum = 0;
for (let temp of temperatures) {
    sum += temp;
}
let meanTemp = sum / temperatures.length;
console.log(`Mean: ${meanTemp}`);
```

With Functions:

javascript

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```
function getMeanTemp(temperatures) {
    let sum = 0;
    for (let temp of temperatures) {
        sum += temp;
    }
    return sum / temperatures.length;
}
```

```
let day1Temps = [12, 13, 14];
```

```
let day2Temps = [17, 16, 15];
```

```
console.log(`Mean Day 1: ${getMeanTemp(day1Temps)}`);
```

```
console.log(`Mean Day 2: ${getMeanTemp(day2Temps)}`);
```

Key Takeaways:

- Functions improve code modularity, reusability, and readability.
- Local variables and the return statement keep code isolated and predictable.
- Parameter validation ensures robust and error-free execution.